

Probability and Statistics III

Lecture Notes

Prof. Josphat Kinyanjui

May 27, 2026

Table of Contents

Multivariate Distribution Theory

We earlier defined a random variable as a real-valued function over a sample space with a probability measure. Many different random variables can be defined over one and the same sample space.

In this section we shall be concerned first with the **bivariate case**, i.e. situations where we are interested at the same time in a pair of random variables defined over a joint sample space. Later we shall extend this discussion to the **multivariate case**, covering any finite number of random variables.

If X and Y are discrete random variables, we write the probability that X will take on the value x and Y will take on the value y as $f(x, y)$. Thus, $f(x, y)$ is the probability of the intersection of the events $X = x$ and $Y = y$.

Joint Probability Distribution (Discrete Case)

Example 1

Two caplets are selected at random from a bottle containing three aspirins, two sedatives, and four laxative caplets.

If X and Y are, respectively, the numbers of aspirin and sedative caplets included among the two caplets drawn, find the probabilities associated with all possible pairs of values of X and Y .

Solution:

The possible pairs are (x, y) where $x \in \{0, 1, 2\}$ and $y \in \{0, 1, 2\}$, subject to $x + y \leq 2$.

For example, the probability associated with $(1,0)$: we want one aspirin, zero sedatives, and hence one laxative. The number of ways is $\binom{3}{1} \binom{2}{0} \binom{4}{1}$ and the total number of ways to select two caplets from nine is $\binom{9}{2}$.

Continuing this way, we obtain:

	x = 0	x = 1	x = 2
y = 0	$f(0,0)$	$f(1,0)$	$f(2,0)$
y = 1	$f(0,1)$	$f(1,1)$	0
y = 2	$f(0,2)$	0	0

It is generally preferable to represent such probabilities by means of a formula — a function $f(x, y)$ defined for any pair of values within the range of X and Y .

Definition 1 – Joint Probability Distribution Function (Discrete)

If X and Y are discrete random variables, the function given by $f(x, y) = P(X = x, Y = y)$ for each pair of values (x, y) within the range of X and Y is called the **joint probability distribution function** of X and Y .

Theorem 1

A bivariate function can serve as the joint probability distribution function of a pair of discrete random variables X and Y if and only if its values $f(x, y)$ satisfy the conditions:

1. $f(x, y) \geq 0$, for each pair of values (x, y) within its domain.
2. $\sum_x \sum_y f(x, y) = 1$, where the double summation extends over all possible pairs within the domain.

Example 2

Determine the value of k for which the function $f(x, y) = k(x + y)$ for $x = 0, 1, 2$ and $y = 0, 1, 2$ can serve as a joint probability distribution function.

Solution:

Substituting the various values of x and y and applying Theorem 1, we get:

$$\sum_x \sum_y k(x + y) = 1$$

Solving for k gives the appropriate value.

Joint Distribution Function (Discrete Case)

Definition 2 – Joint CDF (Discrete)

If X and Y are discrete random variables, the function given by

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t)$$

for $-\infty < x, y < \infty$, where $f(s, t)$ is the value of the joint probability distribution of X and Y at (s, t) , is called the **joint distribution function** or the **joint Cumulative Distribution Function (CDF)** of X and Y .

Example 3

With reference to Example 1, find $F(1, 1)$.

Solution:

As in the univariate case, the joint distribution function is defined for all real numbers. For example, $F(1, 1) = \sum_{x \leq 1} \sum_{y \leq 1} f(x, y)$.

Joint Probability Density (Continuous Case)

Definition 3 – Joint PDF (Continuous)

A bivariate function with values $f(x, y)$, defined over the xy -plane, is called a **joint probability density function** of a continuous random variable X and Y if and only if

$$P[(X, Y) \in A] = \iint_A f(x, y) dx dy$$

for any region A in the xy -plane.

Theorem 2

A bivariate function can serve as a joint probability density function of a pair of continuous random variables X and Y if and only if its values $f(x, y)$ satisfy the conditions:

1. $f(x, y) \geq 0$ for all (x, y)

$$2. \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Example 4

Given the joint probability density function $f(x, y)$ of two random variables X and Y , find $P[(X, Y) \in A]$ where A is a given region.

Solution:

$$P[(X, Y) \in A] = \iint_A f(x, y) dx dy$$

Joint CDF (Continuous Case)

Definition 4 – Joint CDF (Continuous)

If X and Y are continuous random variables, the function given by

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) dt ds, \quad -\infty < x, y < \infty$$

where $f(s, t)$ is the value of the joint probability density of X and Y at (s, t) , is called the **joint Cumulative Density Function** of X and Y .

Analogous to the univariate case, partial differentiation yields:

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y)$$

wherever these partial derivatives exist.

Example 5

If the joint probability density of X and Y is given by $f(x, y)$, find the cumulative density function.

Solution:

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) dt ds$$

Example 6

Find the joint probability density of the two random variables X and Y whose joint cumulative density function is given. Also use the joint probability density function to determine the required probability.

Solution:

(a) Since $f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$, partial differentiation yields the joint PDF.

(b) The probability is evaluated by integrating $f(x, y)$ over the relevant region.

For two random variables, the joint probability density is geometrically a surface, and the probability is given by the volume under this surface over the region A .

Multivariate Case

All the definitions of this section can be generalized to the **multivariate case** with n random variables. Corresponding to Definition 1, the values of the joint probability distribution of n discrete random variables $X_1, X_2, \dots, X_n$ are given by

$$f(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

for each n -tuple $(x_1, \dots, x_n)$ within the range of the random variables; and their joint distribution function is

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Example 7

If the joint probability distribution of three random variables X, Y , and Z is given, find the specified probability.

Solution:

$$P(X = x, Y = y, Z = z) = \sum_x \sum_y \sum_z f(x, y, z)$$

In the continuous case, probabilities are obtained by integrating the joint probability density, and the joint CDF is given by

$$F(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(t_1, \dots, t_n) dt_n \cdots dt_1$$

Example 8

If the trivariate probability density of (X, Y, Z) is given by $f(x, y, z)$, find the probability over a specified region A .

Solution:

$$P[(X, Y, Z) \in A] = \iiint_A f(x, y, z) dz dy dx$$

##Exercises

1. Given the joint probability distribution table of X and Y (with rows $y = 0, 1, 2, 3$ and columns $x = 0, 1, 2$), find the specified probabilities.
2. If the joint probability density of X and Y is given, find the value of c , then find:
 - (a) $P(X < 1, Y < 1)$
 - (b) $P(X + Y < 1)$
 - (c) $P(X > Y)$
3. Determine k so that a given function can serve as a joint probability density function.
4. Find $F(x, y)$ if the joint probability distribution of X and Y is given.
5. Find k if a given bivariate function can serve as a joint probability density.
6. Verify the joint distribution function of a given example.
7. If a trivariate probability density $f(x_1, x_2, x_3)$ is given, find the specified probability.
8. A certain college gives attitude tests in the sciences and humanities. If X and Y are the proportions of correct answers in the two subjects, find the probabilities that a student will get:
 - (a) Less than $\frac{1}{2}$ on both tests.
 - (b) More than $\frac{1}{2}$ on the science test and less than $\frac{1}{4}$ on the humanities test.
9. Suppose P , the price of a commodity (in dollars), and S , its total sales (in units), are random variables with a given joint probability density. Find the probabilities that:
 - (a) The price will be less than 30 cents and sales will exceed 10,000 units.
 - (b) The price will be between 25 cents and 30 cents and sales will be less than 6,000 units.

Marginal and Conditional Distributions

Marginal Distributions

Definition 5 – Marginal Distribution (Discrete)

If X and Y are discrete random variables and $f(x, y)$ is the value of their joint probability distribution at (x, y) , the function given by

$$g(x) = \sum_y f(x, y), \quad \text{for each } x \text{ within the range of } X$$

is called the **marginal probability distribution function** of X .

Correspondingly, the function given by

$$h(y) = \sum_x f(x, y), \quad \text{for each } y \text{ within the range of } Y$$

is called the **marginal probability distribution function** of Y .

Example 9

The joint probability distribution function of random variables X and Y is given in the table below ($x, y \in \{1, 2, 3\}$). Determine the marginal probability distribution functions of X and Y .

	x = 1	x = 2	x = 3	h(y)
y = 1	$f(1,1)$	$f(2,1)$	$f(3,1)$	
y = 2	$f(1,2)$	$f(2,2)$	$f(3,2)$	
y = 3	$f(1,3)$	$f(2,3)$	$f(3,3)$	
g(x)				

Solution:

- (i) The marginal probability distribution function of X : $g(x) = \sum_y f(x, y)$
- (ii) The marginal probability distribution function of Y : $h(y) = \sum_x f(x, y)$

Definition 6 – Marginal Density (Continuous)

If X and Y are continuous random variables and $f(x, y)$ is the value of their joint probability density at (x, y) , the function given by

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

is called the **marginal probability density function** of X .

Correspondingly,

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

is called the **marginal probability density function** of Y .

Example 10

Given the joint probability density $f(x, y)$, find the marginal probability density functions of X and Y .

Solution:

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Joint Marginal Distributions (Multivariate)

When dealing with more than two random variables, we can speak of **joint marginal distributions** of several of the random variables.

If the joint probability of discrete random variables $X_1, X_2, \dots, X_n$ has the value $f(x_1, x_2, \dots, x_n)$, the marginal probability distribution function of X_1 alone is:

$$f_1(x_1) = \sum_{x_2} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

The joint marginal probability distribution function of X_1 and X_2 is:

$$f_{12}(x_1, x_2) = \sum_{x_3} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

For the **continuous case**, summations are replaced by integrals:

$$f_1(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \dots, x_n) dx_2 \cdots dx_n$$

Example 11

Considering the trivariate probability density $f(x_1, x_2, x_3)$, find:

- (i) The joint marginal probability density function of X_1 and X_2 .
- (ii) The marginal probability density function of X_1 alone.

Solution:

$$(i) f_{12}(x_1, x_2) = \int_{-\infty}^{\infty} f(x_1, x_2, x_3) dx_3$$

$$(ii) f_1(x_1) = \int_{-\infty}^{\infty} f_{12}(x_1, x_2) dx_2$$

Conditional Distributions

We define the conditional probability of event A given event B as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0$$

Suppose A is the event $\{X = x\}$ and B is the event $\{Y = y\}$. Then:

$$P(X = x|Y = y) = \frac{f(x, y)}{h(y)}, \quad \text{provided } h(y) > 0$$

Definition 7 – Conditional Distribution (Discrete)

If $f(x, y)$ is the value of the joint probability distribution of the discrete random variables X and Y at (x, y) , and $h(y)$ is the value of the marginal distribution of Y at y , the function given by

$$f(x|y) = \frac{f(x, y)}{h(y)}, \quad \text{for each } x \text{ within the range of } X$$

is called the **conditional distribution of X given $Y = y$** .

Correspondingly, if $g(x)$ is the value of the marginal distribution of X at x :

$$f(y|x) = \frac{f(x, y)}{g(x)}, \quad \text{for each } y \text{ within the range of } Y$$

is called the **conditional distribution of Y given $X = x$** .

Definition 8 – Conditional Density (Continuous)

If $f(x, y)$ is the value of the joint density of continuous random variables X and Y at (x, y) , and $h(y)$ is the value of the marginal density of Y at y , the function given by

$$f(x|y) = \frac{f(x, y)}{h(y)}, \quad \text{for } h(y) > 0$$

is called the **conditional density of X given $Y = y$** .

Correspondingly:

$$f(y|x) = \frac{f(x, y)}{g(x)}, \quad \text{for } g(x) > 0$$

is called the **conditional density of Y given $X = x$** .

Example 12

Given the joint probability density $f(x, y)$, find the marginal densities of X and Y and the conditional density of X given $Y = y$.

Solution:

$$g(x) = \int f(x, y) dy, \quad h(y) = \int f(x, y) dx$$

$$f(x|y) = \frac{f(x, y)}{h(y)}$$

Example 13

With reference to Example 1, find:

- (i) The conditional distribution of X given $Y = 1$.
- (ii) The conditional distribution of Y given $X = 0$.

Solution:

The joint distribution together with marginal totals:

	x = 0	x = 1	x = 2	h(y)
y = 0	$f(0,0)$	$f(1,0)$	$f(2,0)$	
y = 1	$f(0,1)$	$f(1,1)$	0	
y = 2	$f(0,2)$	0	0	
g(x)				

(i) Conditional distribution of X given $Y = 1$: $f(x|1) = \frac{f(x,1)}{h(1)}$

(ii) Conditional distribution of Y given $X = 0$: $f(y|0) = \frac{f(0,y)}{g(0)}$

Independence of Random Variables

When we deal with two or more random variables, questions of **independence** are usually of great importance.

If the values of the conditional distribution of X given $Y = y$ do not depend on y , then $f(x|y) = g(x)$, and the formula yields:

$$f(x, y) = g(x) \cdot h(y)$$

That is, the values of the joint distribution are given by the products of the corresponding marginal distributions.

Definition 9 – Independence (Discrete)

If $f(x_1, x_2, \dots, x_n)$ is the value of the joint probability distribution of n discrete random variables $X_1, \dots, X_n$ at $(x_1, \dots, x_n)$, and $f_i(x_i)$ is the value of the marginal distribution of X_i , then the n random variables are **independent** if and only if

$$f(x_1, \dots, x_n) = f_1(x_1) \cdot f_2(x_2) \cdots f_n(x_n)$$

for all $(x_1, \dots, x_n)$ within their range.

Definition 10 – Independence (Continuous)

If $f(x_1, x_2, \dots, x_n)$ is the value of the joint probability density function of n continuous random variables $X_1, \dots, X_n$ at $(x_1, \dots, x_n)$, and $f_i(x_i)$ is the value of the marginal density of X_i , then the n random variables are **independent** if and only if

$$f(x_1, \dots, x_n) = f_1(x_1) \cdot f_2(x_2) \cdots f_n(x_n)$$

for all $(x_1, \dots, x_n)$ within their range.

Example 14

If the joint probability density of X and Y is given, find:

- (a) The marginal density of X .
- (b) The marginal density of Y .

(c) Whether the two random variables are independent.

Solution:

Compute $g(x) = \int f(x, y) dy$ and $h(y) = \int f(x, y) dx$. Then check whether $f(x, y) = g(x) \cdot h(y)$.

If $f(x, y) \neq g(x) \cdot h(y)$, the two random variables are **not independent**.

Example 15

With reference to Example 1, determine whether X and Y are independent.

Solution:

Using the marginal distributions obtained previously, check whether $f(x, y) = g(x) \cdot h(y)$ for all pairs. If any pair fails, X and Y are **not independent**.

Example 16

Considering independent flips of a balanced coin, let X_i be the number of heads (0 or 1) obtained on the i -th flip. Find the joint probability distribution of these n random variables.

Solution:

Since each X_i has the distribution $P(X_i = x) = \frac{1}{2}$ for $x = 0, 1$, and the random variables are independent, their joint probability distribution is:

$$f(x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{2} = \left(\frac{1}{2}\right)^n, \quad x_i \in \{0, 1\}$$

Example 17

Given independent random variables X_1, X_2, X_3 with probability densities $f_1(x_1), f_2(x_2), f_3(x_3)$, find their joint probability density and use it to evaluate a specified probability.

Solution:

The joint probability density function is:

$$f(x_1, x_2, x_3) = f_1(x_1) \cdot f_2(x_2) \cdot f_3(x_3)$$

Exercises

1. Given the joint probability distribution table of X and Y (with $x \in \{-1, 1\}$ and $y \in \{-1, 0, 1\}$), find:
 - (a) Marginal distribution of X .
 - (b) Marginal distribution of Y .
 - (c) Conditional distribution of X given $Y = 1$.
 - (d) Conditional distribution of Y given $X = 0$.
2. Given the joint probability distribution $f(x, y, z)$, find:
 - The joint marginal distribution of X and Y .
 - The joint marginal distribution of X and Z .
 - The marginal distribution of X .
 - The conditional distribution of Z given $X = x$ and $Y = y$.
 - The joint conditional distribution of Y and Z given $X = x$.
3. Given the joint probability distribution table of X and Y , find:
 - (a) Marginal distribution of X .
 - (b) Marginal distribution of Y .
 - (c) Conditional distribution of X given $Y = -1$.
4. Check whether X and Y are independent for given joint distributions.
5. If the joint probability density of X and Y is given, find:
 - The marginal density of X .
 - The conditional density of Y given $X = x$.
6. If X is the proportion of persons responding to one kind of mail-order solicitation and Y to another, find the probabilities that:
 - At least 30% will respond to the first kind.
 - At most 50% will respond to the second kind given a 20% response to the first.
7. If X is the money (in dollars) a salesperson spends on gasoline per day and Y is the corresponding reimbursement, find:
 - The marginal density of X .
 - The conditional density of Y given $X = x$.
 - The probability that the salesperson will be reimbursed at least \$8 when spending \$12.

8. The useful life (in hours) of a vacuum tube is a random variable with probability density $f(x)$. If three tubes operate independently, find:

- (a) The joint probability density of their lifetimes (X_1, X_2, X_3) .
 - (b) The value of $P(X_1 < X_2 < X_3)$ (or as specified).
-

The Bivariate Normal Distribution

Among multivariate densities of special importance is the **multivariate normal distribution**, which is a generalization of the normal distribution in one variable.

Definition and Properties

Definition 1 – Bivariate Normal Distribution

A pair of random variables X and Y have a **bivariate normal distribution** and are referred to as **jointly normally distributed** random variables, if and only if their joint probability density is given by:

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right] \right\}$$

where $-1 < \rho < 1$.

To study this joint distribution, we first show that the parameters μ_1, μ_2 are the means and σ_1, σ_2 are the standard deviations of X and Y respectively.

Integrating on y from $-\infty$ to ∞ gives the **marginal density of X** . After completing the square and simplifying:

$$g(x) = \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}$$

which is a **normal distribution** with mean μ_1 and standard deviation σ_1 . By symmetry, the marginal density of Y is a normal distribution with mean μ_2 and standard deviation σ_2 .

The parameter ρ is called the **correlation coefficient**, and it can be shown that $-1 < \rho < 1$. It measures how the two random variables X and Y vary together.

Conditional Distributions

Theorem 1 – Conditional Distributions of the Bivariate Normal

If X and Y have a bivariate normal distribution, the **conditional density of Y given $X = x$** is a normal distribution with:

- **Mean:** $\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(x - \mu_1)$
- **Variance:** $\sigma_2^2(1 - \rho^2)$

And the **conditional density of X given $Y = y$** is a normal distribution with:

- **Mean:** $\mu_1 + \rho \frac{\sigma_1}{\sigma_2}(y - \mu_2)$
- **Variance:** $\sigma_1^2(1 - \rho^2)$

Proof:

Writing $f(y|x) = \frac{f(x,y)}{g(x)}$ and simplifying the exponent, one arrives at a normal density with the stated mean and variance. The corresponding result for the conditional density of X given $Y = y$ follows by symmetry.

Key Theorems

Theorem 2 – Independence in the Bivariate Normal

If two random variables have a bivariate normal distribution, they are **independent if and only if $\rho = 0$** .

Proof: (exercise)

Remark: When $\rho = 0$, the random variables are said to be **uncorrelated**. Note that the marginal distributions may both be normal without the joint distribution being bivariate normal — the converse does not necessarily hold.

Examples

Example 1

In a certain population of married couples, the height X (in meters) of the husband and the height Y of the wife have a bivariate normal distribution with given parameters.

- (i) Find the mean and the variance of the height for the husbands and wives.

- (ii) Find the expected height and variance of the husband whose wife is expected to be 1.55 m tall.

Solution:

- (i) Comparing the given density with the general formula, we identify:

$$\mu_1 = \dots, \quad \sigma_1 = \dots, \quad \mu_2 = \dots, \quad \sigma_2 = \dots, \quad \rho = \dots$$

- (ii) By Theorem 1, the conditional mean of X given $Y = 1.55$ is:

$$E(X|Y = 1.55) = \mu_1 + \rho \frac{\sigma_1}{\sigma_2} (1.55 - \mu_2)$$

The conditional variance of X given $Y = y$ is:

$$\text{Var}(X|Y = y) = \sigma_1^2 (1 - \rho^2)$$

Example 2

Let X and Y have a bivariate normal distribution with parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$.

Find:

- (i) $P(\dots)$ [2 marks]
- (ii) $P(\dots|X = 74)$ [6 marks]

Solution:

- (i) Compute using the standard normal CDF.
- (ii) The conditional distribution of Y given $X = 74$ is normal with:

$$\text{Mean} = \mu_2 + \rho \frac{\sigma_2}{\sigma_1} (74 - \mu_1), \quad \text{Variance} = \sigma_2^2 (1 - \rho^2)$$

Hence, standardize and use the standard normal table to find the required probability.

Exercises

1. If the exponent of e of a bivariate normal density is given, find:
 - (a) The parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$.
 - (b) The required probability.
2. If the exponent of e of a bivariate normal density is given, find $P(Y < y_0|X = x_0)$.

3. If X and Y have a bivariate normal distribution with specified parameters, find $P(\dots)$.
4. If U and V have a bivariate normal distribution, and $U = aX + b$, $V = cY + d$, find an expression for the correlation coefficient of U and V .
5. If X and Y have a bivariate normal distribution, their joint moment-generating function is given by:

$$M(t_1, t_2) = \exp \left(\mu_1 t_1 + \mu_2 t_2 + 1/2 \sigma_1^2 t_1^2 + \rho \sigma_1 \sigma_2 t_1 t_2 + 1/2 \sigma_2^2 t_2^2 \right)$$

Verify that: - The first partial derivative with respect to t_1 at $t_1 = t_2 = 0$ equals μ_1 .
 - The second partial derivative with respect to t_1^2 at $t_1 = t_2 = 0$ equals $\mu_1^2 + \sigma_1^2$.
 - The second partial derivative with respect to t_1 and t_2 at $t_1 = t_2 = 0$ equals $\mu_1 \mu_2 + \rho \sigma_1 \sigma_2$.

6. Suppose that X and Y , the height and weight of certain animals, have a bivariate normal distribution with $\mu_1 = 22$ inches, $\mu_2 = 15$ pounds, $\sigma_1 = 2$ inches, $\sigma_2 = 3$ pounds, and $\rho = 0.80$. Find:
 - (a) The expected weight of an animal that is 25 inches tall. **[14.5 pounds]**
 - (b) The expected height of an animal that weighs 20 pounds. **[23.625 inches]**

Transformation of Random Variables

We concern ourselves with the problem of finding the probability distributions or densities of **functions of one or more random variables**. Given a set of random variables $X_1, X_2, \dots, X_n$ and their joint probability or density, we are interested in finding the joint probability distribution of some random variable:

$$Y = u(X_1, X_2, \dots, X_n)$$

This means the values of Y are related to those of the X 's by the equation $y = u(x_1, x_2, \dots, x_n)$.

Several methods are available for solving this kind of problem. The ones we discuss are:

1. The **Distribution Function Technique**
 2. The **Transformation Technique**
 3. The **Moment-Generating Function Technique**
-

The Distribution Function Technique

A straightforward method of obtaining the probability density of a function of continuous random variables consists of:

1. First finding its **distribution function**, then
2. Finding its **probability density by differentiation**.

Thus, if $X_1, X_2, \dots, X_n$ are continuous random variables with a given joint probability density, the probability density of $Y = u(X_1, X_2, \dots, X_n)$ is obtained by first determining:

$$F_Y(y) = \Pr(Y \leq y) = \Pr [u(X_1, X_2, \dots, X_n) \leq y]$$

and then differentiating to get:

$$f(y) = \frac{d}{dy} F(y)$$

Example 1

Problem: The probability density of X is given by:

$$f(x) = \begin{cases} 6x(1-x), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the probability density of $Y = X^3$.

Solution:

Letting $G(y)$ denote the distribution function of Y :

$$\begin{aligned} G(y) &= \Pr(Y \leq y) = \Pr(X^3 \leq y) = \Pr (X \leq y^{1/3}) \\ &= \int_0^{y^{1/3}} 6x(1-x) dx = 3y^{2/3} - 2y \end{aligned}$$

Differentiating:

$$g(y) = \frac{d}{dy} G(y) = 2y^{-1/3} - 2 = \frac{2}{3}(y^{-1/3} - 1) \cdot 3 = 2(y^{-1/3} - 1)$$

More precisely:

$$g(y) = \frac{1}{3}(2y^{-1/3} - 2)$$

$$g(y) = \begin{cases} \frac{1}{3}(2y^{-1/3} - 2), & 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Example 2

Problem: If $Y = |X|$, show that:

$$g(y) = \begin{cases} f(y) + f(-y), & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

where $f(x)$ is the probability density of X .

Solution:

For $y > 0$:

$$G(y) = \Pr(Y \leq y) = \Pr(|X| \leq y) = \Pr(-y \leq X \leq y) = F(y) - F(-y)$$

Differentiating:

$$g(y) = \frac{d}{dy}[F(y) - F(-y)] = f(y) \cdot 1 - f(-y) \cdot (-1) = f(y) + f(-y)$$

Since X cannot be negative, $g(y) = 0$ for $y < 0$. Letting $g(0) = 0$:

$$g(y) = \begin{cases} f(y) + f(-y), & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Example 3

Problem: Given the joint density of X_1 and X_2 :

$$f(x_1, x_2) = \begin{cases} 6e^{-x_1-2x_2}, & x_1 > 0, x_2 > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find the probability density of $Y = X_1 + X_2$.

Solution:

Integrating the joint density over the region where $x_1 + x_2 \leq y$:

$$F(y) = \int_0^y \int_0^{y-x_1} 6e^{-x_1-2x_2} dx_2 dx_1$$

After integration:

$$F(y) = 1 - 2e^{-y} + 3e^{-2y} - 2e^{-3y} \quad (\text{after simplification})$$

Differentiating gives the density:

$$f(y) = \begin{cases} 6e^{-2y} - 6e^{-3y}, & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Exercises — Week 4

1. If $f(x) = 2xe^{-x^2}$ for $x > 0$ (0 elsewhere) and $Y = X^2$, find:
 - (a) The distribution function of Y : $G(y) = 1 - e^{-y}$, $y > 0$
 - (b) The probability density of Y : $g(y) = e^{-y}$, $y > 0$
2. If X has an exponential distribution with parameter λ , use the distribution function technique to find the p.d.f. of $Y = \ln X$.
3. If X has the uniform density with $\alpha = 0$ and $\beta = 1$, find the p.d.f. of $Y = \sqrt{X}$:

$$g(y) = \begin{cases} 2y, & 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

4. If the joint p.d.f. of X and Y is $f(x, y) = 4xye^{-(x^2+y^2)}$ for $x, y > 0$ (0 elsewhere) and $Z = X^2 + Y^2$, find:
 - (a) The distribution function of Z
 - (b) The probability density of Z
5. If X_1 and X_2 are independent exponential random variables with parameters λ_1 and λ_2 , find the p.d.f. of $Y = X_1 + X_2$ when:
 - (a) $\lambda_1 \neq \lambda_2$: $f(y) = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2} (e^{-\lambda_2 y} - e^{-\lambda_1 y})$, $y > 0$
 - (b) $\lambda_1 = \lambda_2 = \lambda$: $f(y) = \lambda^2 y e^{-\lambda y}$, $y > 0$
6. With reference to Exercise 5 (when $\lambda_1 \neq \lambda_2$), show that $Z = \frac{X_1}{X_1 + X_2}$ has a uniform density with $\alpha = 0$, $\beta = 1$.
7. If the joint density of X and Y is $f(x, y) = e^{-(x+y)}$ for $x, y > 0$ (0 elsewhere) and $Z = \frac{X+Y}{2}$, find the p.d.f. of Z using the distribution function technique.

8. The percentages of copper and iron in a certain ore are X_1 and X_2 respectively. Their joint density is:

$$f(x_1, x_2) = \begin{cases} 5, & 0 \leq x_1 \leq 1, 0 \leq x_2 \leq \frac{2-x_1}{2} \\ 0, & \text{elsewhere} \end{cases}$$

Find the p.d.f. of $Y = X_1 + X_2$ and $E(Y)$.

$$g(y) = \begin{cases} 11/9 \cdot y/2, & 0 \leq y < 1 \\ 3(2-y)(7-4y)/22, & 1 \leq y \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

Transformation Technique

One Variable

We can determine the probability distribution or density of a function of a random variable **without first finding its distribution function**.

Discrete Case

In the discrete case, when the relationship between the values of X and $Y = u(X)$ is **one-to-one**, we simply make the appropriate substitution.

Example 4

Problem: If X is the number of heads in four tosses of a fair coin, find the probability distribution of:

$$Y = \frac{1}{X+1}$$

Solution:

Using the binomial distribution with $n = 4$, $p = \frac{1}{2}$:

x	0	1	2	3	4
$f(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Applying $y = \frac{1}{x+1}$:

y	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$
$g(y)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Substituting $x = \frac{1}{y} - 1$ into the binomial formula directly:

$$g(y) = \binom{4}{\frac{1}{y} - 1} \left(\frac{1}{2}\right)^4, \quad y = 1, 1/2, 1/3, 1/4, 1/5$$

Note: The probabilities remain unchanged — only the variable changes from X to Y .

Example 5 — Non One-to-One Case

Problem: Find the probability distribution of $Z = (X - 2)^2$ where X is as in Example 4.

Solution:

x	0	1	2	3	4
$(x - 2)^2$	4	1	0	1	4
$= z$					

Solving $x = 2 \pm \sqrt{z}$ and summing probabilities for each z :

$$h(0) = f(2) = \frac{6}{16} = \frac{3}{8}$$

$$h(1) = f(1) + f(3) = \frac{4}{16} + \frac{4}{16} = \frac{8}{16} = \frac{1}{2}$$

$$h(4) = f(0) + f(4) = \frac{1}{16} + \frac{1}{16} = \frac{2}{16} = \frac{1}{8}$$

z	0	1	4
$h(z)$	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{1}{8}$

Continuous Case — Theorem 1

Theorem 1: Let $f(x)$ be the probability density of the continuous random variable X . If $y = u(x)$ is differentiable and either **strictly increasing or strictly decreasing** for all values where $f(x) > 0$, then the

equation $y = u(x)$ can be uniquely solved for $x = w(y)$, and the probability density of $Y = u(X)$ is:

$$g(y) = f[w(y)] \cdot |w'(y)|, \quad \text{provided } u'(x) \neq 0$$

and $g(y) = 0$ elsewhere.

Proof sketch: For the increasing case, $\Pr(a < Y < b) = \Pr(w(a) < X < w(b))$.

Changing variables $x = w(y)$, $dx = w'(y) dy$ in the integral yields $g(y) = f[w(y)] \cdot |w'(y)|$. The decreasing case follows similarly, with the absolute value ensuring positivity. ▫

Example 6

Problem: If X has the exponential distribution $f(x) = e^{-x}$ for $x > 0$ (0 elsewhere), find the p.d.f. of $Y = \sqrt{X}$.

Solution:

The inverse is $x = y^2$, so $w(y) = y^2$ and $w'(y) = 2y$.

$$g(y) = f(y^2) \cdot |2y| = e^{-y^2} \cdot 2y = 2ye^{-y^2}$$

$$g(y) = \begin{cases} 2ye^{-y^2}, & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Example 7 — Cauchy Distribution

Problem: A spinner gives angle θ with uniform density $f(\theta) = \frac{1}{2\pi}$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ (0 elsewhere). Find the density of $X = a \tan \theta$, the abscissa on the x -axis.

Solution:

The inverse is $\theta = \arctan(x/a)$, so:

$$\frac{d\theta}{dx} = \frac{a}{a^2 + x^2}$$

$$g(x) = \frac{1}{2\pi} \cdot \frac{a}{a^2 + x^2} \cdot 2\pi = \frac{a}{\pi(a^2 + x^2)} \quad \text{for } -\infty < x < \infty$$

This is the **Cauchy distribution**.

Several Variables

Motivation

Given the joint distribution of X_1 and X_2 , we want the distribution of $Y = u(X_1, X_2)$.

Strategy: Introduce an auxiliary variable Z (e.g., $Z = X_2$), find the joint density of (Y, Z) , then integrate out Z to obtain the marginal density of Y .

Theorem 2 — Jacobian Transformation

Theorem 2: Let $f(x_1, x_2)$ be the joint probability density of the continuous random variables X_1 and X_2 . If:

$$y_1 = u_1(x_1, x_2) \quad \text{and} \quad y_2 = u_2(x_1, x_2)$$

are partially differentiable and define a **one-to-one transformation**, with inverse $x_i = w_i(y_1, y_2)$, then the joint density of $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ is:

$$g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)] \cdot |J|$$

where the **Jacobian** is:

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

Example 8 — Poisson Sum

Problem: X_1 and X_2 are independent Poisson random variables with parameters λ_1 and λ_2 . Find the distribution of $Y = X_1 + X_2$.

Solution:

Their joint distribution is:

$$f(x_1, x_2) = \frac{e^{-\lambda_1} \lambda_1^{x_1}}{x_1!} \cdot \frac{e^{-\lambda_2} \lambda_2^{x_2}}{x_2!}$$

Setting $y = x_1 + x_2$ (so $x_1 = y - x_2$) and summing over $x_2 = 0, 1, \dots, y$:

$$h(y) = \sum_{x_2=0}^y \frac{e^{-(\lambda_1+\lambda_2)} \lambda_1^{y-x_2} \lambda_2^{x_2}}{(y-x_2)! x_2!} = \frac{e^{-(\lambda_1+\lambda_2)} (\lambda_1 + \lambda_2)^y}{y!}$$

using the binomial expansion $(\lambda_1 + \lambda_2)^y = \sum_{x_2=0}^y \binom{y}{x_2} \lambda_1^{y-x_2} \lambda_2^{x_2}$.

Conclusion: The sum of two independent Poisson random variables with parameters λ_1 and λ_2 is Poisson with parameter $\lambda_1 + \lambda_2$.

Example 9 — Exponential to Beta/Gamma

Problem: Given joint density $f(x_1, x_2) = e^{-(x_1+x_2)}$ for $x_1, x_2 > 0$ (0 elsewhere), find:

- (a) The joint density of $Y_1 = X_1 + X_2$ and $Y_2 = \frac{X_1}{X_1 + X_2}$
- (b) The marginal density of Y_2

Solution (a):

Solving: $x_1 = y_1 y_2$ and $x_2 = y_1(1 - y_2)$.

$$J = \begin{vmatrix} y_2 & y_1 \\ 1 - y_2 & -y_1 \end{vmatrix} = -y_1 y_2 - y_1(1 - y_2) = -y_1$$

$$|J| = y_1$$

$$g(y_1, y_2) = e^{-y_1 y_2 - y_1(1 - y_2)} \cdot y_1 = y_1 e^{-y_1}$$

$$g(y_1, y_2) = \begin{cases} y_1 e^{-y_1}, & y_1 > 0, 0 < y_2 < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Solution (b):

$$h(y_2) = \int_0^\infty y_1 e^{-y_1} dy_1 = \Gamma(2) = 1! = 1$$

$$h(y_2) = \begin{cases} 1, & 0 < y_2 < 1 \\ 0, & \text{elsewhere} \end{cases}$$

This is the **uniform distribution** on $(0, 1)$.

Example 10 — Triangular Distribution

Problem: $f(x_1, x_2) = 1$ for $0 < x_1 < 1, 0 < x_2 < 1$ (0 elsewhere). Find the marginal density of $Y = X_1 + X_2$.

Solution:

Let $Y = X_1 + X_2$ and $Z = X_2$. Then $x_1 = y - z, x_2 = z, |J| = 1$.

The region maps to $z < y < z + 1$, $0 < z < 1$, i.e., $z < y < z + 1$.

Integrating out z :

$$h(y) = \begin{cases} \int_0^y dz = y, & 0 \leq y < 1 \\ \int_{y-1}^1 dz = 2 - y, & 1 \leq y \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

This is the **triangular distribution**.

Example 11 — Three Variables

Problem: $f(x_1, x_2, x_3) = e^{-(x_1+x_2+x_3)}$ for $x_i > 0$ (0 elsewhere). Find the marginal density of $Y_1 = X_1 + X_2 + X_3$.

Solution:

Let $Y_2 = X_2$ and $Y_3 = X_3$. Then $x_1 = y_1 - y_2 - y_3$, $x_2 = y_2$, $x_3 = y_3$. The 3×3 Jacobian equals 1.

$$g(y_1, y_2, y_3) = e^{-y_1}, \quad y_1 > y_2 + y_3, \quad y_2, y_3 > 0$$

Integrating out y_2 and y_3 :

$$h(y_1) = \int_0^{y_1} \int_0^{y_1-y_3} e^{-y_1} dy_2 dy_3 = \frac{1}{2} y_1^2 e^{-y_1}$$

$$h(y_1) = \begin{cases} \frac{1}{2} y_1^2 e^{-y_1}, & y_1 > 0 \\ 0, & \text{elsewhere} \end{cases}$$

This is a **Gamma distribution** with $\alpha = 3$ and $\beta = 1$.

Exercises — Week 5

1. If X has a hypergeometric distribution with $m = 3$, $N = 6$, $n = 2$, find the p.d. of $Z = (X - 1)^2$.

$$h(0) = \frac{3}{5}, \quad h(1) = \frac{2}{5}$$

2. If X has a binomial distribution with $n = 3$, $p = \frac{1}{3}$, find the p.d. of:

○

$$(a) Y = \frac{X}{X+1}: g(0) = \frac{8}{27}, g\left(\frac{1}{2}\right) = \frac{12}{27}, g\left(\frac{2}{3}\right) = \frac{6}{27}, g\left(\frac{3}{4}\right) = \frac{1}{27}$$

○

$$(b) U = (X - 1)^4: g(0) = \frac{12}{27}, g(1) = \frac{14}{27}, g(16) = \frac{1}{27}$$

3. If $X = \ln(Y)$ has a normal distribution with mean μ and std dev σ , find the p.d.f. of Y (the **log-normal distribution**):

$$g(y) = \frac{1}{\sqrt{2\pi}\sigma y} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right), \quad y > 0$$

4. If $f(x) = kx^3(1 - x^2)^6$ for $0 < x < 1$ (0 elsewhere), find the p.d.f. of $Y = X^2$. Show it is a Beta distribution and find k .

The distribution is Beta with $\alpha = 4, \beta = 2; k = 320$.

5. If X has a uniform density with $\alpha = 0, \beta = 1$, show that $Y = -2\ln X$ has a **gamma distribution**. Find its parameters.
6. If X has a uniform density with $\alpha = 0, \beta = 1$, show that $Y = X^{-1/\alpha}$ (for $\alpha > 0$) has the **Pareto distribution**.
7. Consider $X \sim \text{Uniform}(-1, 3)$:

○

$$(a) \text{ Find the p.d.f. of } Y = |X|: g(y) = \frac{1}{4} \text{ for } 0 \leq y < 1; g(y) = \frac{1}{4} \text{ for } 1 \leq y < 3$$

○

$$(b) \text{ Find the p.d.f. of } Z = (X - Y)^2 = (X - |X|)^2.$$

8. If $f(x_1, x_2) = \frac{x_1 x_2}{36}$ for $x_1, x_2 = 1, 2, 3$, find the p.d. of:

○

$$(a) X_1 X_2$$

○

$$(b) X_1 / X_2$$

9. With the same joint distribution as in Exercise 8, find:

○

$$(a) \text{ The joint distribution of } Y_1 = X_1 + X_2 \text{ and } Y_2 = X_1 - X_2$$

○

(b) The marginal distribution of Y_1 : $g(2) = \frac{1}{36}$, $g(3) = \frac{4}{36}$, $g(4) = \frac{10}{36}$,
 $g(5) = \frac{12}{36}$, $g(6) = \frac{9}{36}$

10. If X_1, X_2, X_3 have the multinomial distribution with $n = 2$, $\theta_1 = \frac{1}{4}$, $\theta_2 = \frac{1}{2}$, $\theta_3 = \frac{5}{12}$, find the joint p.d. of $Y_1 = X_1 + X_2$, $Y_2 = X_1 - X_2$, $Y_3 = X_3$.
11. If X_1 and X_2 are independent binomial with parameters (n_1, θ) and (n_2, θ) , show that $Y = X_1 + X_2$ is binomial with parameters $(n_1 + n_2, \theta)$.
12. If X and Y are independent standard normal, show that $Z = X + Y$ is normally distributed. Find the mean and variance. $\mu = 0$, $\sigma^2 = 2$
13. Given $f(x, y) = 12y(1 - x - y)$ for $0 < x < 1$, $0 < y < 1 - x$ (0 elsewhere):
- (a) Find the joint density of $Z = XY$ and $U = Y$
 - (b) Find the marginal density of Z : $h(z) = 6z - 6z^2 - 12z^2$ for $0 < z < 1$
14. Two independent Cauchy random variables X_1 and X_2 with density $f(x) = \frac{1}{\pi(1+x^2)}$:
- (a) Find the joint density of $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$
 - (b) Find the marginal density of Y_1 : $g(y_1) = \frac{1}{\pi(4+y_1^2)} \cdot 4$
15. Given $f(x, y) = 24xy$ for $0 < x < 1$, $0 < y < 1$, $x + y < 1$, find the joint density of $Z = X + Y$ and $W = X$.
16. Let X and Y be independent gamma random variables. Find the joint density of $U = \frac{X}{X+Y}$ and $V = X + Y$, and identify the marginal density of U .
17. The Maxwell–Boltzmann velocity distribution is $f(v) = kv^2 e^{-\beta v^2}$ for $v > 0$. Show that kinetic energy $E = \frac{1}{2} mV^2$ has a **gamma distribution**.
-

Moment-Generating Function Technique

Overview

Moment-generating functions provide an elegant tool for finding the distribution of **linear combinations of independent random variables**.

The method is based on the uniqueness of the MGF: if $M_Y(t) = M_Z(t)$ for all t in some interval, then Y and Z have the same distribution.

Theorem 3 — MGF of a Sum

Theorem 3: If $X_1, X_2, \dots, X_n$ are **independent** random variables and $Y = X_1 + X_2 + \dots + X_n$, then:

$$M_Y(t) = \prod_{i=1}^n M_{X_i}(t)$$

Proof:

$$M_Y(t) = E[e^{tY}] = E[e^{t(X_1 + \dots + X_n)}] = E[e^{tX_1} \dots e^{tX_n}] = \prod_{i=1}^n E[e^{tX_i}] = \prod_{i=1}^n M_{X_i}(t)$$

(using independence). ▀

Example 12 — Sum of Poisson Variables

Problem: $X_1, \dots, X_n$ are independent Poisson with parameters $\lambda_1, \dots, \lambda_n$. Find the distribution of $Y = X_1 + \dots + X_n$.

Solution:

The MGF of a $\text{Poisson}(\lambda_i)$ is $M_{X_i}(t) = e^{\lambda_i(e^t - 1)}$. By Theorem 3:

$$M_Y(t) = \prod_{i=1}^n e^{\lambda_i(e^t - 1)} = e^{(\lambda_1 + \dots + \lambda_n)(e^t - 1)}$$

Conclusion: $Y \sim \text{Poisson}(\lambda_1 + \lambda_2 + \dots + \lambda_n)$.

Example 13 — Sum of Exponential Variables

Problem: $X_1, \dots, X_n$ are i.i.d. $\text{Exponential}(\lambda)$ (equivalently $\text{Gamma}(1, \lambda)$). Find the distribution of $Y = \sum X_i$.

Solution:

The MGF of $\text{Gamma}(\alpha, \beta)$ is $M_X(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha}$. For $\text{Exponential}(\lambda)$:

$$M_{X_i}(t) = \left(1 - \frac{t}{\lambda}\right)^{-1}$$

$$M_Y(t) = \left(1 - \frac{t}{\lambda}\right)^{-n}$$

Conclusion: $Y \sim \text{Gamma}(n, \lambda)$ with p.d.f.:

$$g(y) = \frac{\lambda^n}{\Gamma(n)} y^{n-1} e^{-\lambda y}, \quad y > 0$$

Applied Example — Lake Victoria Fish

Setup: The number of fish caught per hour at Lake Victoria follows $\text{Poisson}(\lambda = 1.6)$.

(a) Four fish in two hours

Let $Y = X_1 + X_2 \sim \text{Poisson}(3.2)$:

$$P(Y = 4) = \frac{e^{-3.2}(3.2)^4}{4!} = \frac{0.04076 \times 104.86}{24} \approx \mathbf{0.1781}$$

(b) At least two fish in three hours

Let $Y = X_1 + X_2 + X_3 \sim \text{Poisson}(4.8)$:

$$P(Y \geq 2) = 1 - P(Y = 0) - P(Y = 1)$$

$$P(Y = 0) = e^{-4.8} \approx 0.00823, \quad P(Y = 1) = 4.8e^{-4.8} \approx 0.03950$$

$$P(Y \geq 2) = 1 - 0.04773 \approx \mathbf{0.9523}$$

(c) At most three fish in four hours

Let $Y = X_1 + X_2 + X_3 + X_4 \sim \text{Poisson}(6.4)$:

$$P(Y \leq 3) = \sum_{y=0}^3 \frac{e^{-6.4}(6.4)^y}{y!}$$

$$= 0.00166 + 0.01063 + 0.03403 + 0.07253 \approx \mathbf{0.1189}$$

Chebyshev's Theorem

Statement

Chebyshev's Theorem: If μ and σ are the mean and standard deviation of a random variable X , then for any $k > 0$:

$$\Pr(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

Equivalently: $\Pr(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$.

This provides a **universal lower bound** on the probability that X lies within k standard deviations of its mean, valid for *any* distribution.

Proof

By definition:

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Split the integral into three regions:

$$\sigma^2 \geq \int_{-\infty}^{\mu - k\sigma} (x - \mu)^2 f(x) dx + \int_{\mu + k\sigma}^{\infty} (x - \mu)^2 f(x) dx$$

Since $(x - \mu)^2 \geq k^2\sigma^2$ in the outer regions:

$$\sigma^2 \geq k^2\sigma^2 \cdot \Pr(|X - \mu| \geq k\sigma)$$

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$\Pr(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \quad \square$$

Example 1 — Beta Distribution

Problem: X has p.d.f. $f(x) = 630x^4(1-x)^4$ for $0 < x < 1$. Compare the exact probability within two standard deviations of the mean to Chebyshev's bound.

Solution:

Comparing to the Beta(α, β) density, we identify $\alpha = 5, \beta = 5$.

$$\mu = E(X) = \frac{\alpha}{\alpha + \beta} = \frac{5}{10} = 0.5$$

$$\sigma^2 = \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{25}{100 \times 11} = \frac{1}{44}$$

$$\sigma = \frac{1}{\sqrt{44}} \approx 0.1508$$

The interval within 2 standard deviations: $[0.5 - 2(0.1508), 0.5 + 2(0.1508)] = [0.20, 0.80]$.

Exact probability:

$$P(0.20 < X < 0.80) = 630 \int_{0.20}^{0.80} x^4 (1-x)^4 dx \approx \mathbf{0.96}$$

Chebyshev bound (with $k = 2$):

$$1 - \frac{1}{k^2} = 1 - \frac{1}{4} = 0.75$$

The exact probability 0.96 is much stronger than Chebyshev's lower bound 0.75.

Example 2 — Uniform Distribution

Problem: $X \sim \text{Uniform}(-3, 3)$. With $k = \frac{3}{2}$, find:

- (a) The exact probability
- (b) The Chebyshev bound

Solution:

$$\mu = \frac{-3 + 3}{2} = 0, \quad \sigma^2 = \frac{(3 - (-3))^2}{12} = \frac{36}{12} = 3, \quad \sigma = \sqrt{3}$$

With $k = \frac{3}{2}$: $k\sigma = \frac{3}{2}\sqrt{3}$

Exact probability:

$$P\left(|X| < \frac{3\sqrt{3}}{2}\right) = P\left(-\frac{3\sqrt{3}}{2} < X < \frac{3\sqrt{3}}{2}\right) = \frac{1}{6} \cdot 3\sqrt{3} \approx 0.866$$

Actually, since $\frac{3\sqrt{3}}{2} \approx 2.598 < 3$:

$$P = \frac{2 \cdot 2.598}{6} = \frac{5.196}{6} \approx \mathbf{0.866}$$

Chebyshev bound:

$$1 - \frac{1}{(3/2)^2} = 1 - \frac{4}{9} = \frac{5}{9} \approx 0.556$$

Exercises — Week 6

1. **MGF Technique:** If X_1 and X_2 are independent binomial with parameters (n_1, θ) and (n_2, θ) , show using MGFs that $Y = X_1 + X_2 \sim \text{Binomial}(n_1 + n_2, \theta)$.
2. If n independent random variables each have $\text{Gamma}(\alpha, \beta)$ distributions, find the MGF of their sum and identify the distribution. **Answer:** $\text{Gamma}(n\alpha, \beta)$.
3. If $X_i \sim N(\mu_i, \sigma_i^2)$ independently, find the MGF of $Y = \sum X_i$ and identify the distribution.

$$Y \sim N \left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2 \right)$$

4. Prove the generalization of Theorem 3: if $Y = a_1X_1 + \cdots + a_nX_n$ with X_i independent, then $M_Y(t) = \prod_{i=1}^n M_{X_i}(a_it)$.
5. A lawyer receives calls on an unlisted number at rate 2.1/half-hour and on a listed number at rate 10.9/half-hour (independent Poisson). Find the probabilities that in half an hour she receives:
 - (a) 14 calls: **0.1021**
 - (b) At most 6 calls: **0.0259**
6. (Repeat of fish problem — see Section @ref(ex-fish) above.)
7. If the time a doctor spends per patient is $\text{Exponential}(\lambda = \frac{1}{9})$ (minutes), find the probability the doctor spends at least 20 minutes with:
 - (a) One patient: $P(X_1 \geq 20) = e^{-20/9} \approx \mathbf{0.1084}$

- - (b) Two patients: $P(Y \geq 20)$ where $Y \sim \text{Gamma}(2, 1/9)$: **0.3492**
 -
 - (c) Three patients: $P(Y \geq 20)$ where $Y \sim \text{Gamma}(3, 1/9)$: **0.6168**
8. **Chebyshev:** Find the smallest k such that $P(|X - \mu| < k\sigma) \geq$:
- - (a) 0.95: $k = \sqrt{20} \approx 4.47$
 -
 - (b) 0.99: $k = \sqrt{100} = 10$
9. The thiamine content in a bread slice has $\mu = 0.260$ mg, $\sigma = 0.005$ mg. By Chebyshev's theorem, between what values must the thiamine content lie for:
- - (a) At least $\frac{35}{36}$ of all slices: $[0.230, 0.290]$ mg
 -
 - (b) At least $\frac{143}{144}$ of all slices: $[0.200, 0.320]$ mg
10. If $E(X) = 3$ and $E(X^2) = \frac{13}{2}$, use Chebyshev's theorem to find a lower bound for $P(2 < X < 8)$. **Answer:** 0.84

End of STA227 Weeks 4–6 Notes

Week 7: Sampling Distribution

Introduction

Statistics concerns itself mainly with conclusions and predictions resulting from chance outcomes that occur in carefully planned experiments or investigations.

In the **finite case**, these chance outcomes constitute a subset, or **sample**, of measurements or observations from a larger set of values called the **population**. In the **continuous case** they are usually identically distributed random variables, whose distribution we refer to as the **population distribution**, or the *infinite population* sampled.

*Not all samples lend themselves to valid generalizations about the populations from which they came. Most methods of inference are based on the assumption that we are dealing with **random samples**.*

Definition 1 – Random Sample

If $X_1, X_2, \dots, X_n$ are **independent and identically distributed** random variables, we say they constitute a **random sample** from the infinite population given by their common distribution.

If $f(x_1, x_2, \dots, x_n)$ is the value of the joint distribution, we can write:

$$f(x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i)$$

where $f(x_i)$ is the value of the population distribution at x_i .

Statistical inferences are usually based on **statistics** — random variables that are functions of $X_1, X_2, \dots, X_n$.

Definition 2 – Sample Mean and Sample Variance

If $X_1, X_2, \dots, X_n$ constitute a random sample, then:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad (\text{sample mean})$$

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1} \quad (\text{sample variance})$$

For observed sample data, we calculate $\bar{x}$ and s^2 ; these are values of the corresponding random variables $\bar{X}$ and S^2 .

The Distribution of the Mean

Since statistics are random variables, their values vary from sample to sample. Their distributions are called **sampling distributions**.

Theorem 1 – Mean and Variance of $\bar{X}$

If $X_1, X_2, \dots, X_n$ constitute a random sample from an infinite population with mean μ and variance σ^2 , then:

$$E(\bar{X}) = \mu \quad \text{and} \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Proof:

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \cdot n\mu = \mu$$

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

The **standard error of the mean** is $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$. As n increases, $\sigma_{\bar{X}}$ decreases — larger samples yield $\bar{X}$ values closer to μ .

Theorem 2 – Chebyshev’s Bound for $\bar{X}$

For any positive constant c , the probability that $\bar{X}$ falls between $\mu - c$ and $\mu + c$ is at least:

$$1 - \frac{\sigma^2}{nc^2}$$

As $n \rightarrow \infty$, this probability approaches **1**.

Proof:

From Chebyshev’s theorem, for any random variable with mean μ and standard deviation σ , and any $k > 0$:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

Applying this to $\bar{X}$ (which has standard deviation $\sigma/\sqrt{n}$), set $k\sigma_{\bar{X}} = c$, so $k = \frac{c\sqrt{n}}{\sigma}$:

$$P(|\bar{X} - \mu| < c) \geq 1 - \frac{\sigma^2}{nc^2}$$

This result is known as the **Law of Large Numbers**. ■

Central Limit Theorem

Theorem 3 – Central Limit Theorem (CLT)

If $X_1, X_2, \dots, X_n$ constitute a random sample from an infinite population with mean μ , variance σ^2 , and moment-generating function $M_X(t)$, then the **limiting distribution** of:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

as $n \rightarrow \infty$ is the **standard normal distribution**.

Practical rule: The CLT approximation is used when $n \geq 30$, regardless of the shape of the population.

Important note: The CLT does *not* say the distribution of $\bar{X}$ becomes normal (since $\text{Var}(\bar{X}) \rightarrow 0$). It justifies approximating $\bar{X}$ with a normal having mean μ and variance σ^2/n when n is large.

Example 1 – Soft Drink Vending Machine

A vending machine dispenses drinks with mean $\mu = 200$ ml and standard deviation $\sigma = 15$ ml. Find $P(\bar{X} \geq 204)$ for $n = 36$.

Solution:

$$\begin{aligned}\sigma_{\bar{X}} &= \frac{15}{\sqrt{36}} = 2.5 \\ P(\bar{X} \geq 204) &= P\left(Z \geq \frac{204 - 200}{2.5}\right) = P(Z \geq 1.6) \\ &= 1 - P(Z \leq 1.6) = 1 - 0.9452 = \boxed{0.0548}\end{aligned}$$

Example 2 – Gamma Population

A random sample of size $n = 72$ is taken from:

$$f(x) = \frac{1}{16} x e^{-x/4}, \quad x > 0$$

Use the CLT to find $P(\bar{X} > 9)$.

Solution:

Identifying this as a Gamma distribution with $\alpha = 2, \beta = 4$:

$$E(X) = \alpha\beta = 8, \quad \text{Var}(X) = \alpha\beta^2 = 32$$

$$P(\bar{X} > 9) = P\left(Z > \frac{9 - 8}{\sqrt{32/72}}\right) = P(Z > 1.5) = 1 - 0.9332 = \boxed{0.0668}$$

Theorem 4 – Exact Distribution for Normal Populations

If $\bar{X}$ is the mean of a random sample of size n from a **normal** population with mean μ and variance σ^2 , then the **exact** sampling distribution is:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

(This holds for *any* n , without the CLT approximation.)

Exercises – Week 7

1. A random sample of size $n = 100$ is taken from a population with $\mu = 75$ and $\sigma^2 = 256$.
 - (a) Using Chebyshev's theorem, find $P(67 < \bar{X} < 83)$. **[Answer: 0.96]**
 - (b) Using the CLT, find $P(67 < \bar{X} < 83)$. **[Answer: 0.9999994]**
2. A random sample of size $n = 81$ is taken from a population with $\mu = 128$ and $\sigma = 6.3$. Find $P(\bar{X} \notin (126.6, 129.4))$ using:
 - (a) Chebyshev's theorem. **[At most 0.25]**
 - (b) The CLT. **[0.0456]**
3. A random sample of size 64 from a normal population with $\mu = 51.4, \sigma = 6.8$. Find:
 - (a) $P(\bar{X} > 52.9)$ **[0.0388]**
 - (b) $P(50.5 < \bar{X} < 52.3)$ **[0.7108]**
 - (c) $P(\bar{X} < 50.6)$ **[0.1736]**
4. $n = 225$ from an exponential population with $\theta = 4$. Find $P(\bar{X} > 4.5)$ using the CLT.
5. $n = 200$ from a uniform population with $\alpha = 24, \beta = 48$. Find $P(\bar{X} < 35)$. **[0.0207]**

6. $n = 100$ from a normal population with $\sigma = 25$. Find $P(|\bar{X} - \mu| \geq 3)$. **[0.2302]**
 7. Let $\bar{X}$ be the mean of a random sample of size 100 from a distribution with $\sigma^2 = 50$. Find approximately $P(49 < \bar{X} < 51)$.
 8. Let $f(x) = \frac{1}{x^2}$ for $x \geq 1$. For $n = 72$, find approximately the probability that more than 50 items are less than 3. **[0.267]**
 9. $\bar{X}$ is the mean of a random sample of size 128 from a Gamma($\alpha = 2, \beta = 4$) distribution. Find approximately $P(7 < \bar{X} < 9)$. **[0.954]**
 10. Find the approximate probability that the mean of a sample of size 15 from $f(x) = 3x^2, 0 < x < 1$, lies between $\frac{3}{5}$ and $\frac{4}{5}$. **[0.840]**
-

Week 8: The Chi-Square and t-Distributions

The Chi-Square Distribution

A random variable X has the **Chi-square distribution** with r degrees of freedom if its pdf is:

$$f(x) = \frac{1}{2^{r/2} \Gamma(r/2)} x^{r/2-1} e^{-x/2}, \quad x > 0$$

We write $X \sim \chi_r^2$ (or X is χ_r^2).

MGF of the Chi-Square Distribution

$$M_X(t) = (1 - 2t)^{-r/2}, \quad t < \frac{1}{2}$$

Proof sketch: Substituting $y = x(1 - 2t)/2$ into the integral and using the Gamma function identity $\int_0^\infty y^{r/2-1} e^{-y} dy = \Gamma(r/2)$ gives the result above.

Mean and Variance of the Chi-Square Distribution

Differentiating $M_X(t) = (1 - 2t)^{-r/2}$:

$$E(X) = r \quad \text{and} \quad \text{Var}(X) = 2r$$

Theorem 5 – Square of Standard Normal

If $Z \sim N(0,1)$, then $Z^2 \sim \chi_1^2$.

Proof: The MGF of Z^2 is derived as $(1 - 2t)^{-1/2}$, which is the MGF of χ_1^2 . ■

Theorem 6 – Sum of Squared Standard Normals

If $X_1, X_2, \dots, X_n$ are independent $N(0,1)$ random variables, then:

$$Y = \sum_{i=1}^n X_i^2 \sim \chi_n^2$$

Proof: By Theorem 5 each $X_i^2 \sim \chi_1^2$, and since they are independent:

$$M_Y(t) = \prod_{i=1}^n (1 - 2t)^{-1/2} = (1 - 2t)^{-n/2}$$

which is the MGF of χ_n^2 . ■

Theorem 7 – Reproductive Property

If $X_1, \dots, X_n$ are independent with $X_i \sim \chi_{r_i}^2$, then:

$$Y = \sum_{i=1}^n X_i \sim \chi_{r_1+r_2+\dots+r_n}^2$$

Theorem 8 – Difference of Chi-Square Variables

If X_1 and X_2 are independent, $X_1 \sim \chi_{r_1}^2$, and $X_1 + X_2 \sim \chi_{r_1+r}^2$, then $X_2 \sim \chi_r^2$.

Theorem 9 – Sample Mean and Variance from a Normal Population

If $\bar{X}$ and S^2 are the mean and variance of a random sample of size n from $N(\mu, \sigma^2)$, then:

(a) $\bar{X}$ and S^2 are **independent**.

(b) $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$

Proof of (b): Using the identity:

$$\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2$$

The left side is χ_n^2 (by Theorem 6). The second term on the right is χ_1^2 (by Theorems 4 and 5). By Theorem 8, $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$. ■

Example 2 – Semiconductor Thickness

A manufacturing process is “in control” if $\sigma \leq 0.60$ thousandths of an inch. For $n = 20$, the process is declared “out of control” if $\frac{(n-1)S^2}{\sigma^2} \geq \chi_{0.01, 19}^2 = 36.191$.

With $S = 0.84$, $\sigma = 0.60$:

$$\frac{(n-1)S^2}{\sigma^2} = \frac{19 \times (0.84)^2}{(0.60)^2} = 37.24 > 36.191$$

Conclusion: Reject H_0 ; the process is **out of control**.

The t-Distribution

Although $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ is elegant, in practice σ is usually **unknown** and must be replaced by the sample standard deviation S .

Theorem 10 – Definition of the t-Distribution

If $Y \sim \chi_r^2$ and $Z \sim N(0, 1)$ are independent, then:

$$T = \frac{Z}{\sqrt{Y/r}}$$

has the **t-distribution** with r degrees of freedom, with pdf:

$$f(t) = \frac{\Gamma\left(\frac{r+1}{2}\right)}{\sqrt{r\pi} \Gamma\left(\frac{r}{2}\right)} \left(1 + \frac{t^2}{r}\right)^{-(r+1)/2}, \quad -\infty < t < \infty$$

Originally introduced by **W.S. Gosset** under the pen-name “**Student**” (his employer, a brewery, did not permit employee publications). Hence also known as **Student’s t-distribution**.

Theorem 11 – t-Statistic for a Normal Sample

If $\bar{X}$ and S^2 are the mean and variance of a random sample of size n from $N(\mu, \sigma^2)$, then:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

Proof: By Theorem 9, set $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ and $Y = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$, which are independent. Substituting into Theorem 10:

$$T = \frac{Z}{\sqrt{Y/(n-1)}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{(n-1)S^2/[\sigma^2(n-1)]}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1} \quad \blacksquare$$

Example 3 – Gasoline Consumption

In 16 one-hour test runs, an engine averaged $\bar{x} = 16.4$ gallons with $s = 2.1$ gallons. Test the claim that $\mu = 12.0$ gallons per hour.

Solution:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{16.4 - 12.0}{2.1/\sqrt{16}} = \frac{4.4}{0.525} = 8.38$$

From t-tables: $t_{0.005, 15} = 2.947$.

Since $8.38 > 2.947$, we **reject the claim**. The true average consumption exceeds 12.0 gallons per hour.

Week 9: The F-Distribution

The F-Distribution

The **F-distribution** is named after **Sir Ronald A. Fisher**, one of the most prominent statisticians of the 20th century. It was originally studied as the sampling distribution of the ratio of two independent chi-square random variables, each divided by its degrees of freedom.

Theorem 12 – Definition of the F-Distribution

If $U \sim \chi_{n_1}^2$ and $V \sim \chi_{n_2}^2$ are **independent**, then:

$$F = \frac{U/n_1}{V/n_2}$$

has an **F-distribution** with n_1 and n_2 degrees of freedom, with pdf:

$$g(f) = \frac{\Gamma\left(\frac{n_1 + n_2}{2}\right)}{\Gamma\left(\frac{n_1}{2}\right)\Gamma\left(\frac{n_2}{2}\right)} \left(\frac{n_1}{n_2}\right)^{n_1/2} \frac{f^{n_1/2-1}}{\left(1 + \frac{n_1}{n_2}f\right)^{(n_1+n_2)/2}}, \quad f > 0$$

Proof: Apply the change-of-variable $f = \frac{un_2}{vn_1}$ to the joint density of U and V , then integrate out v using the substitution $w = v \left(1 + \frac{n_1}{n_2}f\right)/2$. ■

Mean and Variance of the F-Distribution

Let $X \sim F(m, n)$. Then:

$$E(X) = \frac{n}{n-2}, \quad n > 2$$

$$\text{Var}(X) = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}, \quad n > 4$$

Proof sketch:

Write $X = \frac{U/m}{V/n}$ where $U \sim \chi_m^2$ and $V \sim \chi_n^2$ are independent.

$$E(X) = \frac{n}{m} E(U) \cdot E\left(\frac{1}{V}\right) = \frac{n}{m} \cdot m \cdot \frac{1}{n-2} = \frac{n}{n-2}$$

since $E(1/V) = 1/(n-2)$ for $V \sim \chi_n^2$ (obtained by substitution in the integral). Similarly, $E(1/V^2) = 1/[(n-2)(n-4)]$ for $n > 4$, which yields the variance formula. ■

Applications of the F-Distribution

The F-distribution arises naturally when **comparing variances** σ_1^2 and σ_2^2 of two normal populations.

Theorem 13 – Ratio of Sample Variances

If S_1^2 and S_2^2 are the variances of **independent** random samples of sizes n_1 and n_2 from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively, then:

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$$

Proof: By Theorem 9, $\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi_{n_1-1}^2$ and $\frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi_{n_2-1}^2$, independently.

Substituting into Theorem 12 gives the result. ■

The F-table gives critical values $f_\alpha(n_1, n_2)$ such that $P(F > f_\alpha) = \alpha$ for specified degrees of freedom.

Exercises – Weeks 8 & 9

1. Use Theorem 9 to show that for random samples of size n from $N(\mu, \sigma^2)$, the sampling distribution of S^2 has mean σ^2 and variance $\frac{2\sigma^4}{n-1}$.
2. Show that if $X_1, X_2, \dots, X_n$ are independent χ_1^2 and $Y_n = X_1 + X_2 + \dots + X_n$, then the limiting distribution of:

$$Z_n = \frac{Y_n - n}{\sqrt{2n}}$$

is $N(0, 1)$ as $n \rightarrow \infty$.

3. Using Exercise 2, show that if $X \sim \chi_n^2$ with large n , then:

$$\frac{X - n}{\sqrt{2n}} \approx N(0, 1)$$

4. Use Exercise 3 to find the approximate probability that a χ^2_{50} random variable exceeds **68.0**.
5. Show that for $n > 2$, the variance of the t -distribution with n degrees of freedom is $\frac{n}{n-2}$. (Hint: substitute $t = \sqrt{\frac{n}{1-u^2}} \cdot \text{sign}(t)$)
6. Verify that if $T \sim t_n$, then $T^2 \sim F(1, n)$.
7. Verify that if $X \sim F(n_1, n_2)$ and $n_2 \rightarrow \infty$, the distribution of $Y = n_1 X$ approaches $\chi^2_{n_1}$.
8. If $X \sim F(n_1, n_2)$, show that $Y = \frac{1}{X} \sim F(n_2, n_1)$.
9. Verify that if Y has a Beta distribution with $\alpha = \frac{n_1}{2}$ and $\beta = \frac{n_2}{2}$, then:

$$X = \frac{n_2 Y}{n_1(1 - Y)} \sim F(n_1, n_2)$$

End of STA227 Weeks 7–9 Notes

Week 10: Order Statistics

Introduction

Consider a random sample of size n from an infinite population with a **continuous density**, and suppose we arrange the values $X_1, X_2, \dots, X_n$ according to size.

- The **smallest** value is the random variable Y_1
- The **second smallest** is Y_2
- $\vdots$
- The **largest** is Y_n

These Y 's are called **order statistics**. In particular, Y_r is called the r^{th} order statistic.

We restrict to infinite populations with continuous densities so there is zero probability that any two sample values are equal.

Case $n = 2$

$$y_1 = x_1, y_2 = x_2 \quad \text{when } x_1 < x_2$$

$$y_1 = x_2, y_2 = x_1 \quad \text{when } x_2 < x_1$$

Case $n = 3$

Condition $y_1 \quad y_2 \quad y_3$

$$x_1 < x_2 < x_3 \quad x_1 \quad x_2 \quad x_3$$

$$x_1 < x_3 < x_2 \quad x_1 \quad x_3 \quad x_2$$

$$x_2 < x_1 < x_3 \quad x_2 \quad x_1 \quad x_3$$

Theorem 14 – pdf of the r^{th} Order Statistic

For random samples of size n from an infinite population with pdf $f(x)$, the probability density of the r^{th} order statistic Y_r is:

$$g_r(y_r) = \frac{n!}{(r-1)! (n-r)!} \left[\int_{-\infty}^{y_r} f(x) dx \right]^{r-1} f(y_r) \left[\int_{y_r}^{\infty} f(x) dx \right]^{n-r}$$

Proof:

Divide the real line into three intervals:

1. $(-\infty, y_r)$
2. $[y_r, y_r + h)$ for small $h > 0$
3. $[y_r + h, \infty)$

By the **multinomial distribution**, the probability that $r-1$ sample values fall in interval 1, exactly one falls in interval 2, and $n-r$ fall in interval 3 is:

$$\frac{n!}{(r-1)! 1! (n-r)!} \left[\int_{-\infty}^{y_r} f(x) dx \right]^{r-1} \left[\int_{y_r}^{y_r+h} f(x) dx \right] \left[\int_{y_r+h}^{\infty} f(x) dx \right]^{n-r}$$

By the **law of the mean** from calculus:

$$\int_{y_r}^{y_r+h} f(x) dx \approx f(y_r) \cdot h$$

Dividing by h and letting $h \rightarrow 0$ yields the formula above. ■

Special Cases

Smallest Value Y_1

$$g_1(y_1) = n f(y_1) \left[\int_{y_1}^{\infty} f(x) dx \right]^{n-1}$$

Largest Value Y_n

$$g_n(y_n) = n f(y_n) \left[\int_{-\infty}^{y_n} f(x) dx \right]^{n-1}$$

Sample Median (odd $n = 2m + 1$)

The sample median $\tilde{X} = Y_{m+1}$, with sampling distribution:

$$h(\tilde{x}) = \frac{(2m+1)!}{m! m!} \left[\int_{-\infty}^{\tilde{x}} f(x) dx \right]^m f(\tilde{x}) \left[\int_{\tilde{x}}^{\infty} f(x) dx \right]^m$$

For samples of size $n = 2m$, the median is defined as $\frac{Y_m + Y_{m+1}}{2}$.

Example 4 – Exponential Population

Show that for random samples of size n from an **exponential** population with parameter θ :

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0$$

the sampling distributions of Y_1 , Y_n , and the median are as derived below.

Distribution of Y_1 (Minimum)

$$\int_{y_1}^{\infty} f(x) dx = \int_{y_1}^{\infty} \frac{1}{\theta} e^{-x/\theta} dx = e^{-y_1/\theta}$$

Substituting into the formula for g_1 :

$$g_1(y_1) = n \cdot \frac{1}{\theta} e^{-y_1/\theta} \cdot (e^{-y_1/\theta})^{n-1} = \frac{n}{\theta} e^{-ny_1/\theta}$$

$$g_1(y_1) = \frac{n}{\theta} e^{-ny_1/\theta}, \quad y_1 > 0$$

This is an **exponential distribution** with parameter θ/n .

Distribution of Y_n (Maximum)

$$\int_0^{y_n} f(x) dx = 1 - e^{-y_n/\theta}$$

Substituting into the formula for g_n :

$$g_n(y_n) = n \cdot \frac{1}{\theta} e^{-y_n/\theta} \cdot (1 - e^{-y_n/\theta})^{n-1}$$

$$g_n(y_n) = \frac{n}{\theta} e^{-y_n/\theta} (1 - e^{-y_n/\theta})^{n-1}, \quad y_n > 0$$

Distribution of the Median ($n = 2m + 1$)

Computing the required integrals:

$$\int_{-\infty}^{\tilde{x}} f(x) dx = 1 - e^{-\tilde{x}/\theta}, \quad \int_{\tilde{x}}^{\infty} f(x) dx = e^{-\tilde{x}/\theta}$$

Substituting:

$$h(\tilde{x}) = \frac{(2m+1)!}{m! m!} \cdot (1 - e^{-\tilde{x}/\theta})^m \cdot \frac{1}{\theta} e^{-\tilde{x}/\theta} \cdot (e^{-\tilde{x}/\theta})^m$$

$$h(\tilde{x}) = \frac{(2m+1)!}{m! m!} \frac{1}{\theta} e^{-(m+1)\tilde{x}/\theta} (1 - e^{-\tilde{x}/\theta})^m, \quad \tilde{x} > 0$$

Theorem 15 – Asymptotic Distribution of the Median

For large n , the sampling distribution of the median for random samples of size $2n + 1$ is approximately **normal** with:

$$\text{Mean} = \tilde{\mu}, \quad \text{Variance} = \frac{1}{8n [f(\tilde{\mu})]^2}$$

where $\tilde{\mu}$ is the population median satisfying $\int_{-\infty}^{\tilde{\mu}} f(x) dx = 1/2$.

Comparison with the Mean (Normal Population):

For samples of size $2n + 1$ from a normal population, $\tilde{\mu} = \mu$ and $f(\tilde{\mu}) = \frac{1}{\sigma\sqrt{2\pi}}$, giving:

Statistic	Variance
Mean $\bar{X}$	$\frac{\sigma^2}{2n + 1}$
Median $\tilde{X}$	$\frac{\pi\sigma^2}{4n} \approx \frac{\sigma^2}{2n} \cdot \frac{\pi}{2}$

Conclusion: For large samples from normal populations, the **mean is more reliable** than the median — it has smaller variance (less sampling variability).

Exercises – Week 10

1. Find the sampling distributions of Y_1 and Y_n for random samples of size n from a **continuous uniform** population on $[0, 1]$.
2. Find the sampling distribution of the median for random samples of size $2m + 1$ from the population of Exercise 1.

$$h(\tilde{x}) = \frac{(2m + 1)!}{m! m!} \tilde{x}^m (1 - \tilde{x})^m, \quad 0 < \tilde{x} < 1$$

3. Find the **mean** and **variance** of Y_1 for random samples of size n from the population of Exercise 1.
4. Find the sampling distributions of Y_1 and Y_n for random samples of size n from a **Beta distribution** with $\alpha = 3$, $\beta = 2$.

$$g_1(y_1) = 12n y_1^2 (1 - y_1) (1 - 4y_1^3 + 3y_1^4)^{n-1}, \quad 0 < y_1 < 1$$

5. Find the sampling distribution of the median for random samples of size $2m + 1$ from the population of Exercise 4.
6. Show that the **joint density** of Y_1 and Y_n is:

$$g(y_1, y_n) = n(n - 1) f(y_1) f(y_n) \left[\int_{y_1}^{y_n} f(x) dx \right]^{n-2}, \quad y_1 < y_n$$

(a) Use this to find the joint density of Y_1 and Y_n for an exponential population.

(b) Use this to find the joint density of Y_1 and Y_n for the uniform population of Exercise 1.

7. Using part (b) of Exercise 6, find $\text{Cov}(Y_1, Y_n)$.

$$\text{Cov}(Y_1, Y_n) = \frac{1}{(n+1)^2(n+2)}$$

8. Use the joint density of Y_1 and Y_n from Exercise 6 and the **transformation technique** to find an expression for the joint density of Y_1 and the sample range $R = Y_n - Y_1$.

9. Using Exercise 8 and part (a) of Exercise 6, find the sampling distribution of R for random samples of size n from an **exponential population**.

$$f(R) = \frac{n(n-1)}{\theta^2} e^{-R/\theta} (1 - e^{-R/\theta})^{n-2} \cdot \frac{1}{n-1}, \quad R > 0$$

10. Use Exercise 8 to find the sampling distribution of R for random samples of size n from the **continuous uniform** population of Exercise 1.

11. Use the result of Exercise 10 to find the **mean** and **variance** of R :

$$E(R) = \frac{n-1}{n+1}, \quad \text{Var}(R) = \frac{2(n-1)}{(n+1)^2(n+2)}$$

Week 11: Characteristic Functions

Introduction

In more advanced settings, the **moment-generating function** (MGF) is not always used because many distributions do not have MGFs. Instead, we use the **characteristic function**, which exists for every distribution.

Let i denote the imaginary unit and t an arbitrary real number.

Definition – Characteristic Function

If X is a random variable, its **characteristic function** is $\varphi_X: \mathbb{R} \rightarrow \mathbb{C}$, defined as:

$$\varphi_X(t) = E(e^{itX}) = \begin{cases} \sum_x e^{itx} P(X = x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{itx} f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

This expectation **exists for all** t (since $|e^{itx}| = 1$).

Properties of Characteristic Functions

1. Uniqueness

X and Y have the same characteristic function **if and only if** they have the same distribution function.

2. Moment Generation

If $E(X^n)$ exists, then the n^{th} derivative of $\varphi_X(t)$ exists and:

$$\varphi_X^{(n)}(t) = i^n E(X^n e^{itX})$$

In particular, at $t = 0$:

$$E(X^k) = \frac{\varphi_X^{(k)}(0)}{i^k}$$

Proof:

$$\frac{d}{dt} \varphi_X(t) = \frac{d}{dt} E(e^{itX}) = E\left(\frac{d}{dt} e^{itX}\right) = E(iX e^{itX}) = i E(X e^{itX})$$

$$\frac{d^2}{dt^2} \varphi_X(t) = i \frac{d}{dt} E(X e^{itX}) = i E(X^2 e^{itX}) = i^2 E(X^2 e^{itX})$$

By induction, $\varphi_X^{(n)}(t) = i^n E(X^n e^{itX})$. ■

3. Taylor Expansion

If $E(X^n)$ exists:

$$\varphi_X(t) = \sum_{j=0}^n \frac{(it)^j}{j!} E(X^j) + o_n(t)$$

where $o_n(t)$ satisfies $\lim_{t \rightarrow 0} \frac{o_n(t)}{t^n} = 0$.

Mean and Variance via Characteristic Functions

From Property 2, evaluating derivatives at $t = 0$:

$$E(X) = \frac{\varphi_X'(0)}{i}$$

$$\text{Var}(X) = \frac{\varphi_X''(0)}{i^2} - \left[\frac{\varphi_X'(0)}{i} \right]^2 = -\varphi_X''(0) - [E(X)]^2$$

Connection to MGF: For distributions that have an MGF $M_X(t)$, we have $\varphi_X(t) = M_X(it)$.

Example 1 – Binomial Distribution

Let $X \sim \text{Binomial}(n, p)$ with $q = 1 - p$.

(a) Characteristic Function

$$\begin{aligned}\varphi_X(t) &= E(e^{itX}) = \sum_{x=0}^n e^{itx} \binom{n}{x} p^x q^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (pe^{it})^x q^{n-x} = (q + pe^{it})^n\end{aligned}$$

by the **Binomial Theorem** with $a = q$ and $b = pe^{it}$.

$$\varphi_X(t) = (q + pe^{it})^n$$

(b) Mean and Variance

First derivative:

$$\varphi_X'(t) = n(q + pe^{it})^{n-1} \cdot pie^{it}$$

At $t = 0$, $q + p = 1$:

$$\varphi_X'(0) = npi \implies E(X) = \frac{npi}{i} = np$$

Second derivative (using the product rule):

$$\varphi_X''(t) = n(n-1)(q + pe^{it})^{n-2}(pie^{it})^2 + n(q + pe^{it})^{n-1} \cdot pi^2 e^{it}$$

At $t = 0$:

$$\varphi_X''(0) = n(n-1)p^2 i^2 + npi^2 = -n(n-1)p^2 - np$$

Variance:

$$\begin{aligned}\text{Var}(X) &= \frac{\varphi_X''(0)}{i^2} - [E(X)]^2 = \frac{-n(n-1)p^2 - np}{-1} - n^2 p^2 \\ &= n(n-1)p^2 + np - n^2 p^2 = np - np^2 = np(1-p) = npq\end{aligned}$$

$E(X) = np,$	$\text{Var}(X) = npq$
--------------	-----------------------

Example 2 – Exponential Distribution

Let $X \sim \text{Exponential}(\theta)$ with $f(x) = \frac{1}{\theta} e^{-x/\theta}, x > 0$.

(a) Characteristic Function

$$\begin{aligned}\varphi_X(t) &= \int_0^\infty e^{itx} \cdot \frac{1}{\theta} e^{-x/\theta} dx = \frac{1}{\theta} \int_0^\infty e^{-x(1/\theta - it)} dx \\ &= \frac{1}{\theta} \cdot \frac{1}{1/\theta - it} = \frac{1}{1 - i\theta t}\end{aligned}$$

$\varphi_X(t) = \frac{1}{1 - i\theta t}$
--

(b) Mean and Variance

First derivative:

$$\varphi_X'(t) = \frac{i\theta}{(1 - i\theta t)^2}$$

At $t = 0$: $\varphi_X'(0) = i\theta$, so:

$$E(X) = \frac{i\theta}{i} = \theta$$

Second derivative:

$$\varphi_X''(t) = \frac{2(i\theta)^2}{(1 - i\theta t)^3} = \frac{-2\theta^2}{(1 - i\theta t)^3}$$

At $t = 0$: $\varphi_X''(0) = -2\theta^2$, so:

$$\text{Var}(X) = \frac{-2\theta^2}{i^2} - \theta^2 = 2\theta^2 - \theta^2 = \theta^2$$

$$\boxed{E(X) = \theta, \quad \text{Var}(X) = \theta^2}$$

Exercises – Week 11

For each distribution below, **(a)** find the characteristic function, and **(b)** derive the mean and variance.

1. Discrete Uniform — $P(X = x) = \frac{1}{N}$ for $x = 0, 1, \dots, N - 1$.

$$\varphi_X(t) = \frac{1}{N} \cdot \frac{1 - e^{iNt}}{1 - e^{it}}, \quad E(X) = \frac{N-1}{2}, \quad \text{Var}(X) = \frac{N^2-1}{12}$$

2. Poisson — $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ for $x = 0, 1, 2, \dots$

$$\varphi_X(t) = e^{\lambda(e^{it}-1)}, \quad E(X) = \lambda, \quad \text{Var}(X) = \lambda$$

3. Continuous Uniform — $f(x) = \frac{1}{b-a}$ for $a < x < b$.

$$\varphi_X(t) = \frac{e^{ibt} - e^{iat}}{it(b-a)}, \quad E(X) = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

4. Chi-Squared — $X \sim \chi_n^2$.

$$\varphi_X(t) = (1 - 2it)^{-n/2}, \quad E(X) = n, \quad \text{Var}(X) = 2n$$

5. Normal — $X \sim N(\mu, \sigma^2)$.

$$\varphi_X(t) = e^{i\mu t - \sigma^2 t^2/2}, \quad E(X) = \mu, \quad \text{Var}(X) = \sigma^2$$